Where Does the Drunkard Sleep?

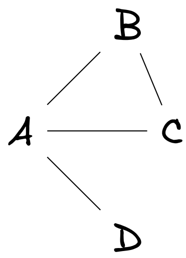
A walker with no map is dropped on a tiny network. At every intersection it picks one of the streets in front of it at random and takes a step. It never remembers where it came from.

Question 1: Before you compute anything, guess. After a very long time, which node do you think the walker is found at most often, and why? ________________

The rest of this sheet checks your guess with a pencil.

Part 1: Transition Probabilities

1. Consider the following undirected network. Represent the network with an adjacency matrix A.



2. A random walk traverses the network by randomly moving from one node to another. Fill the following table with the transition probabilities.

P	To A	To B	To C	To D
From A				
From B				
From C				
From D				

Part 2: Multiple Steps

3. If we start at node A, what is the probability of being at node C after exactly two steps? Show your calculation. Hint: First, calculate the probability $P(j|t=1)$ of being at each node j after one step. Then, multiply the probability of moving from each node j to node i, and sum up the probabilities. Namely,

$$P(i|t=2) = \sum_j \underbrace{P(i|j)}_{\substack{\text{Transition probability} \\ \text{from node } j \text{ to node } i}} \underbrace{P(j|t=1)}_{\substack{\text{Probability of being at} \\ \text{node } j \text{ after 1 step}}}$$

4. If we start at node A, what is the probability of being at node B after exactly three? Show your calculation.

5. How would you calculate the probability of being at any node after T steps, starting from node A? (You don't need to do the calculation, just describe the process using matrix multiplication.)

Part 3: Stationary Distribution

6. Starting from node A, work out the probability of being at each node after 1 step, 2 steps, 3 steps, and so on — keep going until you can see that the numbers have stopped changing. Shade each cell so that darker means higher probability. Twelve columns is plenty; if you need fewer, stop earlier and say why.

step	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												

7. Let's create another heatmap using a random walk starting from node D.

8. Based on your heatmap, what observations can you make about how the probabilities change as the number of steps increases?

9. What makes the stationary probability higher for some nodes than others?